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Essentials of Data Science Laboratory - 2304102L - 2026

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2.1.1. List operations

26:06

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:

- **Add**: Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
- **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
- **Display**: Displays the current list of integers. If the list is empty, display "List is empty".
- **Quit**: Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice".Ensure that the program continues to prompt the user until they choose to quit (option 4).

Sample Test Cases +

Explorer

listOps.py Submit

```python
list = []
while True:
    print("1. Add")
    print("2. Remove")
    print("3. Display")
    print("4. Quit")
    ch = int(input("Enter choice: "))
    if ch==1 :
        num=int(input("Integer: "))
        list.append(num)
        print("List after adding:",list)
```

Debugger

Average time	Maximum time
0.048 s	0.077 s
48.00 ms	77.00 ms

2 out of 2 shown test case(s) p

1 out of 1 hidden test case(s) p

Test case 1 36 ms Debug

Expected output	Actual output
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 11	Integer: 11

Terminal Test cases

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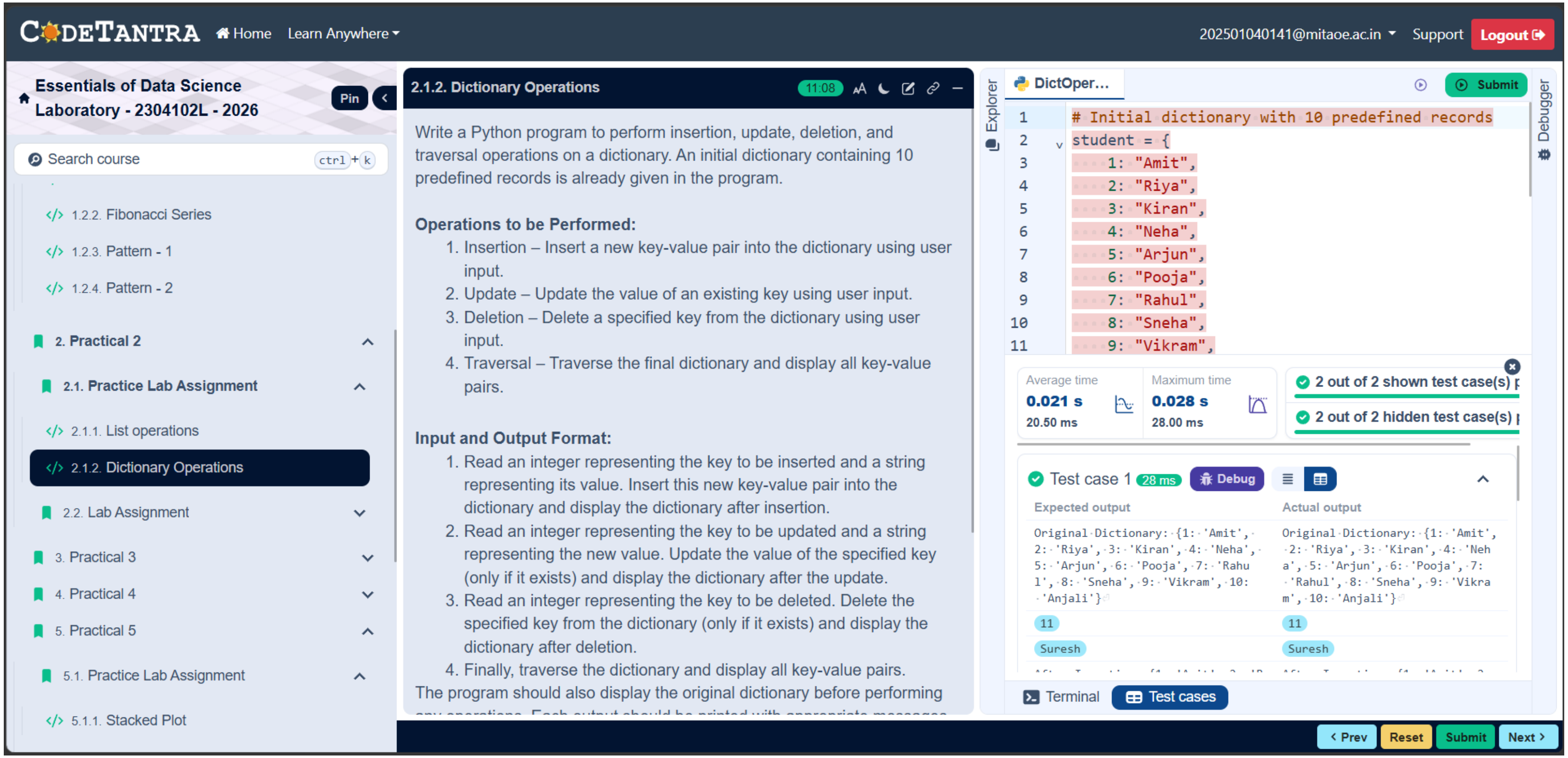
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1.2.2. Fibonacci Series
1.2.3. Pattern - 1
1.2.4. Pattern - 2
2. Practical 2
2.1. Practice Lab Assignment
2.1.1. List operations
2.1.2. Dictionary Operations
2.2. Lab Assignment
3. Practical 3
4. Practical 4
5. Practical 5
5.1. Practice Lab Assignment
5.1.1. Stacked Plot
2.1.2. Dictionary Operations
11:08
Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.
Operations to be Performed:
1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.
Input and Output Format:
1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.
The program should also display the original dictionary before performing
Explorer
DictOper...
Submit
# Initial dictionary with 10 predefined records
student = {
1: "Amit",
2: "Riya",
3: "Kiran",
4: "Neha",
5: "Arjun",
6: "Pooja",
7: "Rahul",
8: "Sneha",
9: "Vikram",
Debugger
Average time
0.021 s
20.50 ms
Maximum time
0.028 s
28.00 ms
2 out of 2 shown test case(s)
2 out of 2 hidden test case(s)
Test case 1
28 ms
Debug
Expected output
Actual output
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}
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11
Suresh
Suresh
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2.2.1. Linear search Technique

03:25

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index. If the element found multiple times, print the starting index
- If the element is not found, print **Not found**.

Sample Test Case:

Input:

1 2 3 4 3 5 6

3

Output:

2

Sample Test Cases +

Explorer

CTP1709... Submit

```python
arr=list(map(int,input().split()))
key=int(input())
found=False
for i in range (len(arr)):
    if arr[i]==key:
        print(i)
        found=True
        break
if not found:
    print("Not found")
```

Debugger

Average time	Maximum time
0.007 s	0.008 s
6.75 ms	8.00 ms

2 out of 2 shown test case(s) p

2 out of 2 hidden test case(s) p

Test case 1 8 ms Debug

Expected output	Actual output
1 2 3 4 3 5 6	1 2 3 4 3 5 6
3	3
2	2

Test case 2 8 ms

Terminal Test cases

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2.2.2. Captain of the Team

01:04

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:
The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format
The output should be the height (in centimeters) of the tallest player.

Sample Test Cases +

captainof... Submit

```python
height = list(map(int,input().split()))
tallest=max(height)
print(tallest)
```

Average time 0.005 s 5.00 ms

Maximum time 0.008 s 8.00 ms

1 out of 1 shown test case(s) p

2 out of 2 hidden test case(s) p

Test case 1 8 ms Debug

Expected output	Actual output
171 169 185 156 174 191 186 190 187 172 160	171 169 185 156 174 191 186 190 187 172 160
191	191

Terminal Test cases

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